

Date	16 July 2026
Project Name	Smart Lender AI: An Intelligent Machine Learning System for Loan Approval Prediction
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Maximum Marks	3 Marks

Requirement Analysis: Data Flow Diagram (Level 1)

The Level 1 Data Flow Diagram (DFD) maps the secure transformation pathways of applicant-submitted data within the Smart Lender AI pipeline. It traces input boundaries starting from external entity interaction, moving through localized form validations, database parameters extraction, machine learning vector transformation, and culminating in structural JSON REST API predictions.

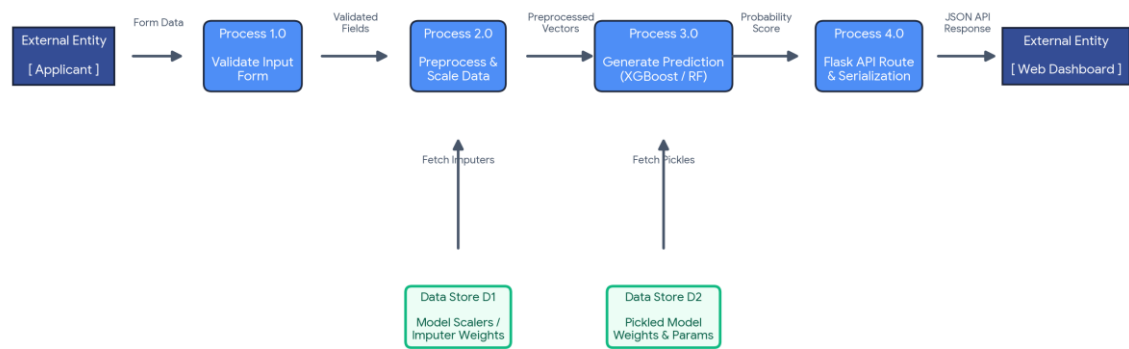


Figure 1: Smart Lender AI Level 1 Data Flow Diagram (DFD)

Process Explanations & Data Transformations

- **Process 1.0 (Validate Input Form):** Consumes raw multi-type variables supplied by the Applicant. It enforces boundary checks (e.g., verifying that numeric fields contain positive values and that co-applicant income contains zero or positive floats), and blocks malicious inputs prior to serialization.
- **Process 2.0 (Preprocess & Scale Data):** Reads system imputation and standard scaling specifications dynamically from Data Store D1. Imputes missing categorical and numerical entries (e.g., employing mode and mean models) and standardizes numerical dimensions to keep the system scaling consistent with training boundaries.
- **Process 3.0 (Generate Prediction):** Pulls serialized classifier weights (XGBoost / Random Forest pickles) from Data Store D2. Operates on preprocessed vectors to produce continuous probabilistic ranges, which are mapped to categorical classifications (Approved/Denied).
- **Process 4.0 (Flask API Route & Serialization):** Encapsulates predictions and visual parameter arrays into clean, secure JSON formats. It dynamically streams outputs back to the interactive Web Dashboard in under 1 second.